

Research Article

Development of an IoT-based Monitoring System for Greenhouse Gas Emissions and Soil Health in Paddy Fields

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Abstract.

The agricultural sector significantly contributes to greenhouse gas (GHG) emissions, particularly methane, due to anaerobic conditions in paddy soils. Effective monitoring of these emissions is critical for climate change mitigation and sustainable agricultural practices. This study developed an Internet of Things (IoT)-based system for real-time monitoring of methane emissions and soil health. The system integrated advanced sensors: RS-CH4-X-2-EX for methane detection, CWT-BL-ORP-2K-S for redox potential measurement, and QC-S8 for soil chemical analysis. The design process focused on creating a user-friendly platform capable of providing continuous and accurate data. Field testing evaluated the system's performance under flooded paddy field conditions. The RS-CH4-X-2-EX sensor demonstrated high precision ($\pm 2\%$) and rapid detection (< 10 seconds), accurately capturing methane concentration dynamics. The CWT-BL-ORP-2K-S sensor effectively monitored redox potential with stable readings in anaerobic environments, while the QC-S8 sensor provided detailed information on soil moisture, nutrients, and pH levels, critical for precision agriculture. The sensors were seamlessly integrated into an IoT platform, allowing remote data access and real-time visualization, enhancing decision-making for sustainable farming. This system offers a cost-effective and scalable approach for monitoring GHG emissions and improving soil health, enabling more efficient resource use, increased productivity, and reduced environmental impacts.

Keywords: IoT-based monitoring, methane emissions, paddy fields, soil health

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1. Introduction

Global climate change and global warming have become critical issues worldwide. One of the main causes of climate change is the increase in greenhouse gas (GHG) emissions, including carbon dioxide (CO₂) and methane (CH₄), which mostly originate from human activities such as industry, transportation, and the agricultural sector [1]. Especially in agriculture, rice fields are known to be a major contributor to methane emissions [2]. This is due to the anaerobic conditions in waterlogged soil, which encourage the activity of methane-producing microorganisms [3]. Although many studies have been conducted to study the emissions of this gas, the measurement and monitoring of GHG emissions from agricultural land, especially rice fields, still need to be improved in terms of efficiency and data accuracy.

Manual monitoring of GHG emissions is often inefficient, requires a lot of time, effort, and cost, and can only provide instantaneous data (snap-shots) without continuity. As a result, analysis of long-term emission trends to identify patterns and dynamic changes in GHG emissions is difficult. In recent decades, the development of Internet of Things (IoT)-based technology has offered a promising solution to overcome these limitations. IoT allows real-time and continuous data collection that can be directly accessed through a digital platform for deeper analysis [4]. With this technology, it is possible to carry out more accurate and continuous monitoring, which will provide a more comprehensive picture of the contribution of the agricultural sector to climate change.

This study aims to develop a prototype of an IoT-based GHG emission measuring tool that can be applied to rice fields, especially to detect and record methane and carbon dioxide emissions. The main innovation of this tool is its ability to measure emissions in real-time, enable direct data transmission to a digital platform, and offer continuous data access. In addition, the graphic design of this tool is developed with a user-friendly interface, making it easier for farmers, researchers, and policy makers to understand and utilize the data produced [5].

Previous studies on IoT-based monitoring systems have shown promising results in various agricultural and environmental applications, such as the development of greenhouse monitoring systems to optimize growing conditions [6], prototypes for detecting temperature, humidity, and air quality using integrated IoT and solar technologies [7], and advancements in IoT-enabled smart city technologies for broader environmental management [8]. However, the application of such systems specifically to rice fields

remains underexplored, particularly in the context of prototypes that integrate automated GHG monitoring with user-friendly data interpretation and accessibility. This is particularly evident in the limited number of prototype designs that integrate automated monitoring functions with accessible and user-friendly data interpretation. The novelty of this study lies in the development of a tool that not only provides continuous data on methane and carbon dioxide emissions in real time but also has a graphical interface design that makes it easy to use by stakeholders from various backgrounds. This is expected to encourage wider adoption of IoT technology in monitoring emissions in the agricultural sector and support national and international GHG emission reduction policies. Overall, this prototype is expected to be a practical and affordable solution in monitoring GHG emissions in rice fields [2]. With more accurate and sustainable data, researchers and policymakers can design more precise and data-based climate change mitigation measures, and support global efforts to reduce GHG emissions from the agricultural sector.

2. Materials and Methods

This research employs an experimental approach to design and develop a prototype Internet of Things (IoT)-based methane gas measurement device integrated with a chamber for rice plants. The study begins with an analysis of device requirements based on the observed parameters, including methane concentration and several soil physical and chemical properties. The device design is created using SolidWorks CAD software. The main components of the device are divided into two parts: hardware and IoT. The hardware includes sensors, a microcontroller, and a power supply, while the IoT platform used is ThingSpeak.

The primary methane sensor is the RS-CH4-X-2-EX methane gas detector (Figure 1.a), manufactured by Shandong Jianda Renke Electronic Technology Co., Ltd. (China). This sensor can detect methane concentrations up to 0–100% Lower Explosive Limit (LEL) with a resolution of 1% LEL, providing accurate readings with a precision level of $\pm 5\%$ full scale (FS) and repeatability $\leq 2\%$. The sensor operates on a power supply of 10–30VDC and outputs RS485 signals. Additional sensors include a redox potential sensor and soil physicochemical property sensors. The redox potential sensor, model CWT-BL-ORP-2K-S by ComWinTop Co., Ltd. (China) (Figure 1.b), measures a range of -2000 to 2000 mV with a sensitivity of 1 mV and an accuracy of 6 mV, consuming less than 1W of power and requiring a 12–24V power supply.

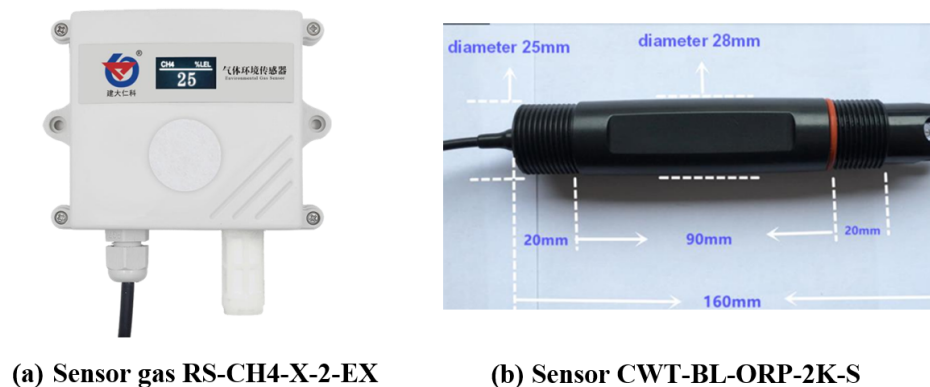


Figure 1: Methane and soil oxidation-reduction potential sensor.

For soil analysis, a QC-S8 soil sensor from Shenzhen Minqin Electronic Technology Co., Ltd. (China) is utilized (Figure 3). This sensor measures variables such as soil temperature (-30 to 70°C , $\pm 0.2^{\circ}\text{C}$ accuracy), moisture (0 – 100% , $\pm 2\%$ accuracy), pH (3 – 10 , ± 0.01 accuracy), electrical conductivity (0 – $2000\ \mu\text{S}/\text{cm}$, $\pm 1\ \mu\text{S}/\text{cm}$ accuracy), salinity (0 – $2000\ \mu\text{S}/\text{cm}$, $\pm 1\ \mu\text{S}/\text{cm}$ accuracy), and NPK levels (0 – $1999\ \text{ppm}$, $\pm 3\ \text{ppm}$ accuracy). The sensor outputs RS485 signals.



Figure 2: Sensor QC-S8.

The device employs an ESP32 microcontroller, supported by various components to ensure optimal functionality. A GSM SIM800 module enables data transmission over the internet, while the RS485 interface facilitates stable long-distance communication with additional sensors. Data is stored on an SD card with a Real-Time Clock (RTC). A 20×4 LCD displays real-time information for user convenience. The device is designed to be portable, powered by a sealed lead-acid battery, but it can also be connected to a direct power source if needed.

The hardware assembly involves integrating the methane sensor, microcontroller, and communication module into a transparent chamber. The software development

uses C++ programming to enable data transmission to the ThingSpeak IoT platform. ThingSpeak is chosen for its cost-free nature and real-time data visualization capabilities. Key features of ThingSpeak include channels for data storage (up to eight fields per channel), API keys for secure access, MATLAB analytics for on-platform data processing, real-time visualization tools, and alert systems for automated notifications based on specific conditions.

The assembled prototype undergoes functional testing, including methane sensor calibration using methane gas. Field testing is conducted by placing the device in a chamber over rice plants cultivated in buckets for four weeks. Methane data collected by the sensor is validated against measurements obtained using Gas Chromatography-Mass Spectrometry (GC-MS) in laboratory analysis. Soil physical and chemical data are compared with standardized laboratory analysis or existing calibrated devices. The collected data are analyzed to evaluate methane concentrations, device accuracy, and the relationship between methane emissions, environmental factors, and plant growth. The prototype's performance is assessed based on accuracy and sensitivity. This study aims to produce a reliable and efficient methane measurement device while providing scientific data relevant to rice cultivation. The research is planned to last six months, encompassing design, development, testing, and data analysis stages.

3. Results and Discussion

The RS-CH₄-X-2-EX sensor demonstrated a high level of accuracy in detecting methane concentrations in paddy rice chambers. Methane, a potent greenhouse gas, is often emitted in rice paddies due to anaerobic conditions in flooded soils. The sensor was able to detect methane levels from as low as a few parts per million (ppm), with a high degree of sensitivity. This sensitivity is essential for real-time monitoring in agricultural systems, especially in paddy fields where methane emissions can fluctuate based on soil moisture and organic matter content. According to the sensor's specifications, its accuracy was within a $\pm 2\%$ tolerance, confirmed by a calibration process using standard reference gases, ensuring precise measurements [9].

The fast response time of the RS-CH₄-X-2-EX sensor was another key feature. The sensor responded to significant changes in methane concentration within 10 seconds, providing near-instantaneous data to facilitate timely decisions in agricultural management. This is particularly beneficial in real-time monitoring scenarios where rapid changes in gas concentrations are expected due to varying irrigation schedules or

fertilization practices. The sensor's ability to maintain stable readings even under changing environmental conditions also enhances its reliability in fluctuating agricultural environments. This characteristic makes the RS-CH₄-X-2-EX suitable for continuous monitoring in paddy rice chambers, where conditions can change rapidly due to water management and soil characteristics [10].

In addition to detecting methane, the RS-CH₄-X-2-EX sensor is immune to interference from other common gases, such as carbon dioxide and nitrogen oxide, which is a critical aspect in agricultural environments. This feature ensures that the sensor specifically detects methane concentrations without being influenced by background gases, making it more reliable for gas emissions monitoring. The semiconductor metal oxide (SMO) sensing technology used by this sensor enables it to perform oxidation-reduction reactions with target gases, resulting in detectable changes in electrical conductivity. This method allows for continuous, high-precision gas monitoring, which is essential for mitigating methane emissions in agriculture, particularly in regions with intensive rice cultivation [4]. Figure 4 presents gas emission measurements with a chamber in a greenhouse.



Figure 3: Measurement of gas emissions in the chamber.

The CWT-BL-ORP-2K-S sensor, used to measure Oxidation-Reduction Potential (ORP), provided essential data for understanding the redox dynamics in flooded agricultural soils. In rice paddies, the oxidation-reduction state of the soil influences the availability of

nutrients and the formation of greenhouse gases such as methane. This sensor detected ORP values within a range of -200 mV to +1000 mV, making it suitable for monitoring flooded soils. The fast response time of less than 5 seconds ensured that the ORP values were recorded accurately and promptly, which is crucial for tracking soil conditions that influence gas emissions [11]. Consistent ORP readings support effective monitoring and management of paddy soils, enabling farmers to adjust irrigation practices to minimize gas emissions.

The CWT-BL-ORP-2K-S sensor also showed minimal deviation in its measurements, with an accuracy of ± 5 mV, making it a reliable tool for continuous monitoring in flooded agricultural systems. The stability of the sensor's readings indicates its potential to provide valuable data for long-term studies of soil health and environmental changes. Redox potential directly impacts microbial activity in soil, which in turn affects methane production in paddy fields. By monitoring the ORP, the sensor allows farmers and researchers to optimize soil management practices that can reduce the emission of greenhouse gases. For example, altering the frequency of water management practices based on ORP data can reduce methane emissions significantly, aligning with sustainable agricultural goals [11].

The QC-S8 soil sensor demonstrated significant utility in precision agriculture, particularly in optimizing soil management practices. This sensor measures critical parameters such as soil moisture, pH, and nutrient levels (Nitrogen, Phosphate, and Potassium), providing real-time data that helps farmers make informed decisions regarding irrigation and fertilization. The ability to monitor soil moisture in particular is essential in paddy fields, where water levels directly impact the microbial processes that influence methane emissions. The QC-S8 sensor's integration into an IoT-based platform enables farmers to receive data remotely, allowing them to monitor soil conditions across large areas without the need for frequent field visits [12].

Through the integration of the QC-S8 sensor with cloud-based platforms, farmers can make more precise decisions about irrigation, fertilization, and crop rotation, ultimately improving agricultural productivity. The data collected by the sensor provides a comprehensive view of the soil's nutrient content, which can help mitigate the overuse of chemical fertilizers that often leads to soil degradation and nutrient imbalances. This real-time data also allows for more sustainable land management practices, ensuring that nutrient application is aligned with the crop's specific needs at any given time. As a result, the QC-S8 sensor plays a crucial role in reducing both environmental impact and operational costs, while simultaneously increasing agricultural yields [13]. Figure 5

presents an illustration of an IoT-based chamber with automatic measurement of soil and greenhouse gas sensors along with its monitoring flow chart. Meanwhile, Figure 6 presents the prototype of the IoT chamber.

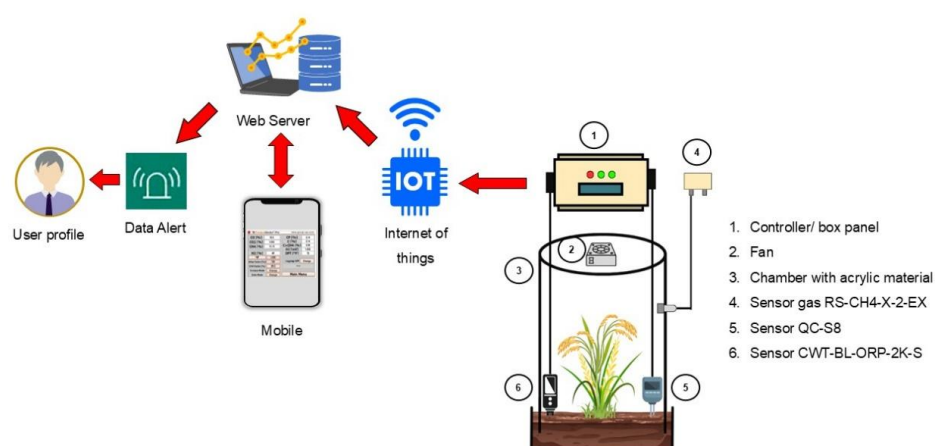


Figure 4: Illustration of IoT chamber with gas and soil sensors.

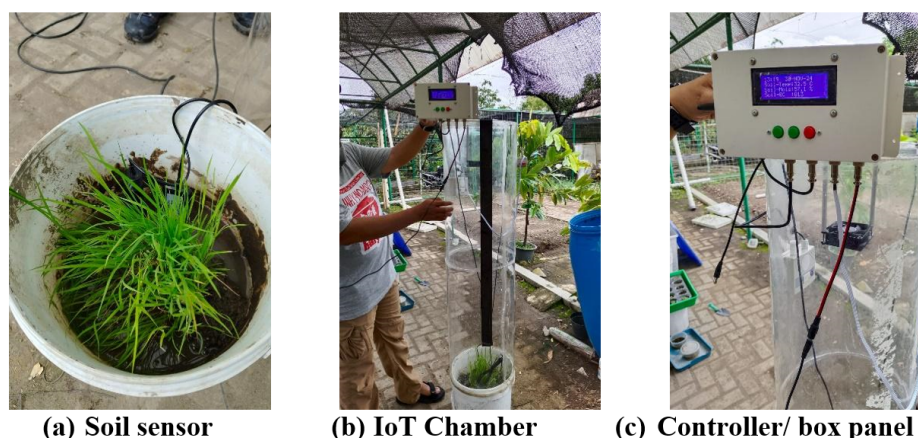


Figure 5: Prototype of IoT chamber with gas and soil sensors.

Another notable benefit of the QC-S8 sensor is its ability to help detect soil degradation, particularly in areas that have been subject to intensive agricultural practices, such as over-fertilization. Soil degradation often leads to reduced agricultural productivity and increased methane emissions. The real-time data provided by the QC-S8 can help farmers identify early signs of soil deterioration, allowing them to take corrective actions before significant damage occurs. By adjusting practices such as irrigation timing, fertilization schedules, and crop rotation based on the sensor's feedback, farmers can improve soil health, reduce emissions, and enhance crop yield sustainably. This makes the QC-S8 sensor an invaluable tool in the broader context of sustainable agriculture and environmental stewardship [2].

Recent research conducted by Zhang et al. [14] developed a WG-IoT-MS system that integrates low-cost sensors with intelligent algorithms to monitor water quality and GHG emissions simultaneously. The WG-IoT-MS system will assess CO₂ and N₂O emissions in various water bodies, including rice paddies and lakes, with the methane monitoring system optimised specifically for flooded rice paddies. The WG-IoT-MS system uses a support vector regression model to predict dissolved N₂O concentrations. Whereas in our research, the IoT-based system for methane and soil health concentrates on methane detection and soil chemistry using sensors. The methane monitoring system we developed integrates RS-CH4-X-2-EX and CWT-BL-ORP-2K-S sensors to measure methane and redox potential with high precision. In addition, we added instruments to assess soil health, including moisture, nutrients and pH, which are critical for precision agriculture. Both approaches emphasise cost-effectiveness, with the WG-IoT-MS system reducing costs by 60% and the methane monitoring system offering a scalable solution for farmers. The ability to remotely access and visualise data in both systems improve decision-making for sustainable farming practices.

4. Conclusion

The integration of advanced sensors such as RS-CH4-X-2-EX, CWT-BL-ORP-2K-S, and QC-S8 in agricultural monitoring systems demonstrates significant potential in improving precision agriculture practices, particularly in managing gas emissions and soil health. The RS-CH4-X-2-EX sensor accurately detects methane concentrations in paddy fields with high sensitivity and rapid response, making it a reliable tool for real-time greenhouse gas monitoring. Its resistance to interference from other gases further enhances its functionality in agricultural environments where emission levels can fluctuate due to varying farming practices. Overall, the combination of these sensors within an IoT-based framework provides a robust solution for addressing key challenges in sustainable agriculture. By enabling precise monitoring of gas emissions and soil health, this approach supports data-driven strategies to mitigate the environmental impact of agricultural activities while maximizing crop yields.

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